

05.1	Draws gradient as a tangent to the curve on the graph ✓ Extracts at least one piece of data from the graph that is used to find a gradient or the gradient of a chord (condoning powers of ten) ✓ 15 to 20 m s⁻² ✓	3	Candidates who use a small triangle can access the second and third marks only. Candidates who use 2 widely spaced points on the curve can access the 2nd mark only
05.2	Evidence of determining the area under the graph ✓ 1 square is equivalent to 2.5×10^{-4} (m) OR 9 to 11 squares ✓ 2.3×10^{-3} to 2.7×10^{-3} (m) ✓	3	Stated or seen on the graph
Total		6	

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
06.1	Uses $s = \frac{1}{2}gt^2$ ✓ 0.585 (s) to at least 2 sf ✓	2	Substitution or rearrangement seen Accept 0.58
06.2	20(.3) (m s ⁻¹) OR 19.8 (m s ⁻¹) when using 0.6 s ✓	1	
06.3	Uses Pythagoras ✓ 49(.4) (m s ⁻¹) ✓ Uses appropriate trig relationship 32° to the horizontal OR Correct scale diagram seen Scale given 0.1 or 0.2 cm per m s ⁻¹ 48 to 50 m s ⁻¹ 30° to 34° to the horizontal	4	Angle must be shown on diagram or clearly stated “to the horizontal” 58° to vertical acceptable Candidates who show an incorrect direction in their vector diagram lose the final mark
Total		7	

Question	Marking guidance	Mark	Comments
	2.5 marks		

Question	Marking guidance	Mark	Comments
08.1	(Horizontal $u =$) $48\cos 15$ seen ($= 46.4 \text{ m s}^{-1}$) ✓ ($D = u\cos 15 \times t = 46.4 \times 3.0 =$) 139 (m) ✓	2	Accept 138 m if 46 m s^{-1} is used.
08.2	$48\sin 15$ leading to 12.4 m s^{-1}	1	Must be given to at least 3sf
08.3	Use of $s = ut + \frac{1}{2}at^2$ (e.g. $(12.4 \times 3.0) + 0.5 \times (-)9.8 \times 3.0^2$) ✓ $= -6.9 \text{ m}$ ✓ height $h = (12 - \text{their } 6.9) = 5.1 \text{ m}$ ✓	3	Allow methods that involve a combination of the other <i>suvat</i> equations Accept any clear correct alternative use of the uniform acceleration equations
Total		6	

Question	Marking guidance	Mark	Comments
			Accent <i>F</i> x's with symbols defined for MP1

16	A
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17	A
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18	B
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Question	Marking guidance	Mark	Comments
01	The object's direction is changing ✓ Acceleration is rate of change of velocity ✓ Velocity is a vector quantity or velocity is speed in a given direction or mention of an appropriate circumstance (eg circular motion)✓	3	Accept statement that an object is accelerating if it's velocity is changing or $a = \frac{\Delta v}{t}$ but not just $a = \frac{v}{t}$ for MP2

05.1	Use of Pythagoras: expect to see $1.1^2 + 0.65^2$ ✓ 1.28 (m s ⁻¹) to at least 3 sf ✓	2	Accept even if there are more than 3 terms in the Pythagoras equation Not 1.27 m s ⁻¹
05.2	Magnitude = 0.25 m s ⁻¹ Direction = (towards the) west	1	Accept just W or "to the left"

Question	Marking guidance	Mark	Comments
08.1	$v_h = \frac{25.0}{0.260}$ or $v_h = 96(.15)$ seen ✓ Use of tan (1) leading to 1.678 to at least 3 sf ✓	2	Use of tan must be seen. Use of sin loses the mark even if the numbers seem to be correct. Accept appropriate combination of cos and sin
08.2	Use of $s = ut + \frac{1}{2}at^2$ ✓ $s = 0.10$ to 0.11 ✓ Between 0.10 m and 0.11 m <u>above</u> the centre of the target ✓	3	Expect to see 0.105 (m) but allow range for different degrees of rounding / use of approximate value for v_h Accept valid approach using different equations of motion Both magnitude and direction must be there for the final mark and the answer must include an appropriate unit

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
09.1	Reasonable tangent drawn at $t = 16 \text{ s}$ ✓ Correct values from tangent used ✓ Deceleration = $0.35 - 0.41 \text{ m s}^{-2}$ ✓	3	Allow intercepts on the t axis of between 22 and 27 Candidates who find the gradient of a triangle around 16 s can access 1 mark max. Tolerate small triangle on a tangent – accept any correct data. Tolerate lack of minus sign
09.2	Attempt to use area ✓ Medium square equivalent to 1.25 m or large square equivalent to 5 m or 11.4 to 12.6 large squares or 45.5 – 50.5 medium squares or finds distance represented by initial triangle and rectangle correctly (= 40 m) ✓ $d = 57 - 63 \text{ m}$ ✓	3	Give equal credit if small squares are used.
09.3	Starts at candidate's answer to 09.2 at time $t = 21 \text{ (s)}$ ✓ Then any 2 from Curve with (increasing) negative gradient from $t = 21 \text{ (s)}$ to $t = 25 \text{ (s)}$ ✓ Straight line with negative gradient from $t = 25 \text{ (s)}$ to $t = 30 \text{ (s)}$ ✓ Additional detail: either: initial curve has negative gradient that gets steeper with time or final displacement is candidate's 09.2 – 14 m ✓	MAX 3	If everything correct but with a positive gradient, candidates may access MP2 and/or MP3. Candidates who plot distance from the point at which the turn is made instead of the distance from the start can access MP2 and MP3 ie MAX 2 Marks available only for: - a curve with (increasing) positive gradient from $t = 21 \text{ (s)}$ to $t = 25 \text{ (s)}$ - a straight line with positive gradient from $t = 25 \text{ (s)}$ to $t = 30 \text{ (s)}$ or final displacement is 14 m

Question	Marking guidance	Mark	Comments
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15	C	40 m s^{-1}
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	P = lift Q = drag R = weight S = thrust (owtte) Any 2 correct gets 1 mark ✓ All correct gets 2 marks ✓	Allow alternatives terms Do not accept 'gravity' for R.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	MAX 3 speed increases or aircraft accelerates (initially)... ✓ air resistance (Q) increases as speed increases ✓ aircraft acquires a new constant speed when air resistance matches increased value of S ✓ aircraft climbs as lift increases (as it depends on the speed of the air over the aerofoil) ✓	In MP2 reference to speed increasing can be seen in MP1.	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	use of resultant force = $9870 - 9600 (= 270)$ (N) ✓ use of $m = \frac{W}{g}$ or 979 (kg) or 980 (kg) ✓ Use of $F = ma$ ✓ $0.276 \text{ (m s}^{-2}\text{) to at least 3 sf}$ m s^{-2} ✓	Need to see a subject for each of MP1, MP2 and MP3 Calculator value is 0.275906 MP4 requires use of 9.81 m s^{-2} for g	4	1 × AO1 2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Use of $v^2 = u^2 + 2as$ ✓ 16.7 m s^{-1} to at least 3 sf ✓	Allow combining suvat equations Must see equation for MP1. Accept 16.5 or 16.6 Allow energy approach.	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.5	Use of Pythagoras ✓ $72 \text{ (m s}^{-1}\text{)} \quad \checkmark$ Use of $\tan^{-1}\left(\frac{16.6}{70}\right) \quad \checkmark$ $13^\circ \quad \checkmark$	Expect to see $70^2 + 16.6^2$ Accept use of tan to find the angle (2 marks) followed by any appropriate trig to find magnitude of resultant velocity (2 marks) Use of the 'show that' value gives 14°	4	1 × AO2 3 × AO3
Total			15	

Question	Answers	Additional comments/Guidelines	Mark	AO
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16	D	The gradient of a distance–time graph is equivalent to speed.	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	calculates u_{vert} or $4.5 \sin 15$ or $4.5 \cos 75$ seen ✓ partially correct substitution into $s = ut + \frac{1}{2}at^2$ ✓ numerically correct substitution ✓ 28 (m) ✓	MP2: any suitable combination of the equations of motion condone inconsistent signs for MP2 and MP3 correct final answer scores all marks condone -28 (m)	4	AO2 × 4

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	horizontal (components of) velocity are the same for both sandbag and balloon ✓ no horizontal resultant force/acceleration on sandbag ✓	condone horizontal speed for horizontal (component of) velocity throughout	2	AO2 × 2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	the idea that it is the gradient at the steepest point ✓ draw a tangent at the steepest point and determine the gradient (of that tangent) ✓	steepest point could be named i.e. 27s, 25m MP2 is dependent on MP1	2	AO1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	correctly reads off displacement at 50 s (= 9 m) ✓ ₁ calculates total distance travelled = 41 m ✓ ₁ ✓ ₂ use of average speed = total distance/total time to give 41/50 = 0.82 (m s ⁻¹) ✓ ₃		3	AO2 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	uses displacement/50 s ✓ ecf from 06.2 towards B /to the right owtte ✓	expect 9/50 = 0.18 (m s ⁻¹) ignore sign for MP1 condone positive or original direction if velocity and speed have been reversed in 06.2 and 06.3 , give ecf for this part	2	AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Attempts to calculate the area under graph ✓ Distance = 0.046 to 0.049 ✓ Time = 19×10^{-3} to 19.5×10^{-3} AND Use of average speed = $\frac{\text{their distance from an area}}{\text{time}}$ ✓	Allow other correct methods Ignore POT in MP3 Their distance must be from a valid method expect to see 2.4 to 2.6 m s^{-1}	3	AO2 × 2 AO3 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Determines a or uses $f = ma$ with their a ✓ 770 N ✓	Expect to see $\frac{3.2}{2.3 \times 10^{-3}} = 1390 \text{ m s}^{-2}$ Accept answer in the range 690 N to 870 N consistent with read-offs for Δt of 2.0 to 2.5 ms Do not penalise POT error if penalised in 04.1	2	AO2 × 1 AO3 × 1
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	$v = (15 \sin 38 =) 9.2 \text{ (9.23) (m s}^{-1}\text{) } \checkmark$		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Correct substitution into equation of uniform motion or conservation of energy $\checkmark$ 4.31 (≥ 3 sf) (m) $\checkmark$	Expect to see $\frac{(0) - 9.2^2}{2 \times -9.81}$ MP2 dependent on MP1 Condone $v = 9.2$ and $a = +9.81$ 4.34 if 9.23 used allow ecf from 08.1 for MP1 only	2	AO1 $\times 1$ AO2 $\times 1$

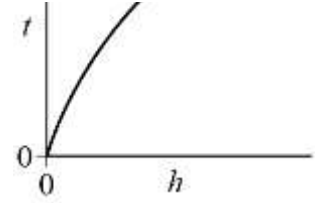
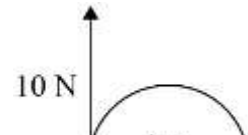
Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	Correct use of an equation of uniform motion $\checkmark$ 0.937 (≥ 3 sf) (s) $\checkmark$	ecf incorrect value from 08.1 or 08.2 MP2 dependent on MP1 Do not allow u and a to have the same sign 0.941 if 4.34 used	2	AO1 $\times 1$ AO2 $\times 1$

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Correct use of equation of uniform motion $s = ut + \frac{1}{2}at^2$ ✓ Calculates time to fall from max height to 2.56 m = 0.60 (0.602) s ✓ 1.5 (1.54) (s) ✓	Use of $s = ut + \frac{1}{2}at^2$ and solving for t can gain full marks	3	AO1 × 1 AO2 × 1 AO3 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.5	Finds horizontal component of velocity OR uses $d = 1.5 \times u$ ✓ 18 (m) ✓	ecf incorrect value from 08.4	2	AO1 × 1 AO2 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.6	Max 2 from: Reaches lower maximum height ✓ Reaches maximum height at an earlier time ✓ The ball will pass under the crossbar or hit the crossbar lower ✓	Condone horizontal distance would be less for MP3	2	AO2 × 1 AO3 × 1
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
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23	A		
			AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Uses $F = ma$ to calculate acceleration of B Or use of $F = \frac{\Delta mv}{\Delta t}$ ✓ Attempts to use $s = ut + \frac{1}{2}at^2$ or other equation of motion to calculate s ✓ 0.56 (m) ✓	Expect to see $\frac{410}{56}$ or 7.3(2) (m s^{-2}) Allow MP2 for candidates who think that $u = 0$ Award both MP1 and MP2 for use of $F = \frac{\Delta mv}{\Delta t}$ and $\Delta \text{KE} = \frac{1}{2}m(v^2 - u^2)$ from energy approach	3	$2 \times \text{AO2}$ $1 \times \text{AO3}$

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Concludes that B has the greater magnitude of acceleration with some justification ✓ Idea that the magnitude of the force on each is the same (Newton's 3rd law) and B has a smaller mass and reference to $a = \frac{F}{m}$ OR Change in momentum is the same for A and B (conservation of momentum with no external forces) and B has a smaller mass plus reference to $\Delta p = m\Delta v$ (and period of the acceleration is the same for both) ✓		2	AO2
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Initial vertical component of the velocity is zero for both balls and vertical acceleration is the same / 9.81 m s^{-2} for both balls ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Use of $v^2 = u^2 + 2as$ to give $h = 0.90 \text{ m}$ ✓	Allow $h = 0.899\dots (\text{m})$	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	Correct rectangle or triangle of velocities with scale quoted ✓ $6.5 \pm 0.2 (\text{m s}^{-1})$ ✓ Scale diagram must use half the available space ✓ $50^\circ \pm 2^\circ (\text{to the vertical})$ ✓	MP2 and MP4 may be awarded for an algebraic method for max 2 marks	4	1 × AO1 1 × AO2 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Plastic ball R will take a longer time to reach the ground than steel ball P because of air resistance ✓ Because $F = ma$ AND any one from ✓ • Air resistance on plastic ball R is greater than on steel ball P (at the same speed) because of plastic ball R's greater diameter • Air resistance has greater effect on time for plastic ball R because of plastic ball R's smaller mass • The resultant force on steel ball P is greater		2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.5	Idea that v_s is smaller because the horizontal deceleration (due to air resistance) is greater ✓ Idea that v_s is smaller because time of fall is longer (due to reduced vertical component) giving longer time for the deceleration to have an effect ✓	allow v_s is smaller because air resistance is greater	2	1 × AO2 1 × AO3
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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18	C	$2.1 \times 10^{-16} \text{ N}$	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Removes variation in reaction time ✓ Time too short to measure with a stopwatch without getting a large uncertainty in the measurement ✓		2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Any two from: ✓✓ • repeat and average • use a greater range of data • use smaller intervals • obtain more data sets		2	AO4

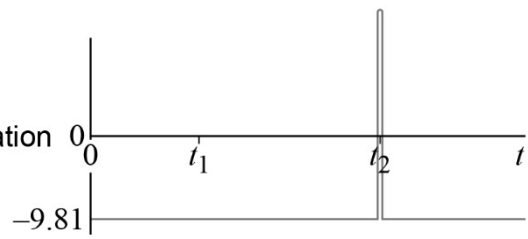
Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	t^2 should be to 3 sf / should not be recorded to 2 sf ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Use $s = \frac{1}{2}gt^2$ and compare with $y = mx + c$ ✓ Plot s against t^2 or $2s$ against t^2 ✓ Determine gradient and multiply by 2 ✓	Accept if axes reversed	3	AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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19	D	velocity	AO1

22	B	downwards with an increasing speed	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Single horizontal line either above or below axis to at least approximately t_2 ✓ −9.8(1) labelled on the y-axis and line drawn horizontally from this point. Allow inverted graph labelled 9.8(1) ✓ Vertical (or almost vertical) line drawn at t_2 that goes through the x-axis ✓	 Accept the bounce to start at t_2 and not go beyond the “t” in “the” from the wording above. Condone a lack of ruler in MP2 and MP3 For 3 marks valid line must continue past t_2	3	2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Any two from: ✓✓ - determines initial vertical velocity ($18\sin 49$ or 13.58 seen) - uses appropriate equation of motion with their vertical velocity - adds 1.5 m to their height 11 (m) ✓	mp1 and mp2 can also be awarded for $mg\Delta h = \frac{1}{2}mv^2 \text{ route}$ expect 9.406 m $g = 9.81$ gives 10.906 m $g = 9.8$ gives 10.916 m	3	2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Minimum velocity is at the peak / maximum height ✓ Horizontal component is constant AND (at peak) vertical velocity = 0 ✓	For MP1, allow correct annotation on Figure 3 Condone it has horizontal speed AND vertical speed = 0 for MP2 For MP2 allow correct energy argument including reference to conservation of energy.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	$18 \cos 49 = 11.8 \text{ (m s}^{-1}\text{)} \checkmark$		1	AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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